

# ARIYUS-ONE

The Quantum Bio-Resonance Vocal Karaoke Studio & DAW

In-Person Presentation Pitch Deck & Technical Manual

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# ARIYUS-ONE: THE QUANTUM BIO-RESONANCE VOCAL STUDIO

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## **Executive Pitch Deck & Comprehensive Feature Index**

*Designed for In-Person Presentations & Print Distribution*

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# 1. Executive Summary

## The Elevator Pitch

**Ariyus-One** is the world's first-of-its-kind **Bio-Resonance Web Audio DAW & Vocal Alignment Platform**. It merges cutting-edge digital signal processing (DSP), real-time vocal biofield telemetry, and gamified alignment challenges to transform singing from standard entertainment into a therapeutic, bio-harmonized art form.

Unlike traditional recording software that relies on artificial, pitch-correcting "Auto-Tune" to mask vocal characteristics, Ariyus-One guides vocalists to align their organic frequencies with natural mathematical constants, such as **Solfeggio scales** and **432 Hz organic cosmic tuning**. It is a fully featured virtual recording studio, voice analysis suite, and community ecosystem built entirely for the modern web browser.

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## 2. The Core Value Proposition

| Traditional Vocal DAWs                                                                                                       | The Ariyus-One Difference                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Artificial Pitch Correction:</b> Forces vocals onto artificial, tempered scales, introducing synthetic digital artifacts. | <b>Organic Frequency Tuning:</b> Uses real-time autocorrelation to guide singers to stabilize their natural pitch at therapeutic hertz ratios.      |
| <b>Flat Stereo Output:</b> Standard left/right panning without spatial or energetic context.                                 | <b>3D Spatial Soundstage:</b> Dynamic interactive coordinates allowing vocalists to pan tracks and position vocal layers in space.                  |
| <b>Basic Visual Waveforms:</b> Decibel meters that show only volume peaks.                                                   | <b>Chakra Biofield Lightfield:</b> Analyzes vocal jitter, shimmer, and HNR to visualize human electromagnetic biofield projections in real-time.    |
| <b>Purely Technical Mixing:</b> Standard EQ, compression, and reverb.                                                        | <b>ARC-5 DSP Effects Rack:</b> Modulates vocal layers using ring modulators, comb resonators, and solfeggio carrier signals for cellular resonance. |

### 3. Technology Stack Architecture

Ariyus-One runs entirely client-side with cloud-synchronized user accounts. This structure guarantees low-latency digital signal processing and secure data storage.



- **Frontend Core:** React.js, Web Audio API, HTML5 Canvas 2D renderers.
- **Backend Infrastructure:** Firebase Authentication (secure login), Firestore Database (user metrics and community feed), Firebase Storage (MP3 recording hosting).
- **Audio Processing Engine:** Custom-built **ARC-5 Web Audio Graph** incorporating biquad filters, delay nodes, waveshaper distortion curves, convolver reverbs, and twin sine oscillators.



## 4. Comprehensive Feature Index

### Feature Node 1: The Home Nexus

The command center of the Ariyus-One experience, displaying real-time profile states and tracking progression.

- **Ariyus Journey Level Tracker:** Tracks user progression using a logarithmic level system (e.g., *Seeker, Resonator, Harmonizer, Alchemist, Luminary*) fueled by XP achievements.
- **Active User Membership Badging:** Displays subscription tiers (*Free, Ariyus Pro, Creator*) with dynamic CSS glow rings.
- **Dynamic CSS Particle Background:** Renders 40 floating cosmic micro-particles moving via random Bezier vectors, mirroring organic cellular patterns.
- **Cosmic Sponsor Terminal:** Contextual banner ads matching the glassmorphic space theme (gated for Free users).

### Feature Node 2: Alignment Challenges Engine

A gamified engine that hooks into the studio and rewards bio-resonance training.

- **Cosmic Breath Challenge:** Demands that the vocalist hold a steady, organic 432 Hz frequency (+/- 5 Hz) continuously for 6.0 seconds. High-precision microphone trackers verify the tone in real-time.
- **Harmonic Alignment Challenge:** Unlocks upon achieving an A+ Pitch Stability rating ( $\geq 90\%$ ) on a finalized performance track.
- **XP Reward Sync:** Syncs completion statuses immediately to Firestore and awards +100 XP / +150 XP.

### Feature Node 3: The Song Library (Sing Console)

The catalogue of target-frequency backing tracks.

- **Solfeggio Categorization:** Tracks are filtered by target frequencies:
  - **396 Hz (Liberation):** UT key for releasing fear and sub-conscious blockages.
  - **417 Hz (Change):** RE key for clearing traumatic patterns.
  - **432 Hz (Cosmic Sync):** Natural harmonic organic tuning.

- **528 Hz (Transformation):** MI key for DNA vitality and repair.
- **639 Hz (Harmonize):** FA key for relationship and interpersonal coherence.
- **741 Hz (Awakening):** SOL key for clearing intuition and expression.
- **852 Hz (Spiritual Order):** LA key for cosmic order alignment.
- **Freestyle Resonance Mode:** Enables recording over a blank cosmic hum without backing instrumentation, allowing for meditation practice.

## Feature Node 4: High-Precision Recording Studio

The core interface where live vocals are captured, analyzed, and calibrated.

- **Autocorrelation Pitch Tracker:** Runs a mathematical time-domain autocorrelation function 10 times per second to isolate the fundamental frequency of the user's voice in Hertz.
- **Cosmic Breath HUD overlay:** Renders a real-time glowing gold progress bar tracking the user's progress toward the 6.0-second sustain limit.
- **Voice-Reactive Visualizer:** Draws high-refresh frequency domain waveform peaks inside an HTML5 canvas.
- **Off-grid Sandbox Simulator:** Includes fallback frequency synthesis that simulates pitch tracking for testing when microphone permissions are denied or in offline demo environments.

## Feature Node 5: The ARC-5 DSP Effects Rack

A proprietary mixing deck that applies real-time acoustic modifications to vocals and backing tracks.

- **Ring Modulator:** Locks the user's voice to the Solfeggio carrier wave sidebands, generating metallic harmonics.
- **Comb Resonator:** Adds a feedback delay loop tuned precisely to the inverse of the Solfeggio hertz key, physically reinforcing the target frequency.
- **Acoustic Coupling:** A dynamic peaking filter that shifts the backing track's frequency range in real-time to match the singer's vocal pitch.
- **Binaural Beating:** Generates a twin-oscillator Solfeggio carrier wave with an 8 Hz offset between the left and right channels to entrain theta brainwaves.
- **Galactic Reverb:** An algorithmic room convolver that passes dry vocals through a randomized soundscape decay buffer (2.5s duration, 2.0 decay exponent).



- **Warm Harmonics:** A tube waveshaper distortion curve that injects warm, even-order overtones.
- **Cyber Chorus:** Modulates a short delay line via a 1.5 Hz LFO to double the presence of the vocals.
- **Vocal Clarity Filter:** A highpass biquad filter cutting out low-frequency microphone mud below 120 Hz.
- **Hyper Bass Boost:** A low-shelf filter adding a warm +8.0 dB boost to the backing track's low frequencies.

## Feature Node 6: 3D Spatial Soundstage

An interactive, tactile coordinates panel mapping spatial depth.

- **2D Node Grid Mapping:** Drag-and-drop nodes represent the *Voice*, *Backing Track*, and *Solfeggio Hum*.
- **Left-to-Right Panning:** Controls the Web Audio StereoPannerNode's panning factor (-1.0 to 1.0) along the X-axis.
- **Front-to-Back Depth:** Adjusts the gain coefficients along the Y-axis, simulating relative distance from the listener.
- **Orbital Binaural Drift:** Swings the Solfeggio hum left and right in a sinusoidal orbit (50ms interval loop) when Binaural Beating is enabled.

## Feature Node 7: Biofield Diagnostics & Visualizers

Extracts biological data from the voice and maps it to ancient energy zones.

- **Voice Biomarkers Parser:** Extracts vocal telemetry:
  - **Jitter (Pitch Variance):** Micro-fluctuations in pitch stability.
  - **Shimmer (Amplitude Variance):** Micro-fluctuations in volume stability.
  - **HNR (Harmonics-to-Noise Ratio):** The ratio of pure harmonic peaks to breathy, background air noise.
  - **Spectral Centroid:** The center of mass of the vocal spectrum, identifying tone brightness.
- **Chakra Biofield Lightfield:** Automatically renders a translucent human silhouette on an HTML5 canvas overlayed with four glowing energy centers:
  - **Third Eye (Ajna):** Tied to pitch stability (Jitter).
  - **Throat (Vishuddha):** Tied to vocal clarity (HNR).

- **Heart (Anahata):** Tied to volume consistency (Shimmer).
- **Root (Muladhara):** Tied to tone depth (Spectral Centroid).
- **Vocal Resonance Prescription:** Diagnoses vocal deficiencies and suggests customized Solfeggio codes and FX combinations to stabilize the voice.
- **AI Voice Coach:** Analyzes biomarkers and prints step-by-step larynx, diaphragm, and resonance exercises (Pro-tier only).

## Feature Node 8: Vector SVG Card Export

- **Aura SVG Generator:** Builds a vector certificate including the singer's vocal identity (e.g., Alto, Tenor), dominant resonance frequency, biomarker stats, chakra alignments, and a secure verification hash.
- **Download Utility:** Triggers an automatic download of a high-resolution, scalable `.svg` image suitable for printing or sharing on social media.

## Feature Node 9: Multi-Track Studio DAW Workstation

A separate workstation environment reserved for paid subscribers that mirrors the features of classic audio editing suites.

- **Multi-DAW Capabilities:** Provides multi-track timeline recording where users can sync vocals with multiple custom-imported audio files.
- **Waveform Segment Editing:** Visualizes audio segments as interactive waveforms that can be split, trimmed, and dragged across the timeline.
- **Automation Envelopes:** Renders volume automation overlay nodes that allow users to draw volume changes directly onto individual audio clips.
- **Sound Forge Mixing & Master Mixdown:** Consolidates all active tracks, volume levels, and panning automation into a single, high-fidelity `.wav` output buffer.

## Feature Node 10: Multiplanetary Collaboration Lobby (Frequency Lab)

- **Multiplayer Synchronizers:** Connects users in a virtual lobby to run bio-resonance sweeps together.
  - **Synchronized Solfeggio Maps:** Displays a live constellation map showing real-time frequency alignment across all connected users.
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## 5. Subscription & Business Model

Ariyus-One utilizes a tiered model powered by integrated payment processors.



### Tier 1: Free Tier (\$0/mo)

- Access to the Home Nexus and basic Sing catalog.
- Basic recording capabilities.

- Access to 4 standard ARC-5 DSP modules (Ring Modulator, Comb Resonator, Vocal Clarity, Hyper Bass).
- Ad-supported.

### **Tier 2: Ariyus Pro (\$19/mo)**

- Ad-free.
- Unlocks all 9 ARC-5 DSP modules (including Galactic Reverb, Cyber Chorus, and Binaural Beating).
- Unlocks full Solfeggio frequency controls.
- Unlocks the AI Voice Coach Analysis.

### **Tier 3: Creator Tier (\$49/mo)**

- Includes all Pro features.
  - Unlocks the Multi-Track DAW Workstation.
  - Allows uploading custom backing tracks and managing guide vocals.
  - Grants access to the Creator Dashboard for performance analytics.
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## 6. Development Status & Roadmap

1. **Phase 1 (Completed):** Core Web Audio API graph and pitch detection algorithms.
2. **Phase 2 (Completed):** UI/UX Design System, 3D Spatial Soundstage, and Chakra canvas engines.
3. **Phase 3 (Completed):** User authentication, profile progression, and the Alignment Challenges Engine.
4. **Phase 4 (In Progress):** Integrating Stripe payment terminals and multi-track mixdown tools.
5. **Phase 5 (Future):** Mobile applications (iOS/Android native wrappers) and VST plugins for integration with standard DAWs.